

Learning from Demonstration for Complex Tasks: Vision-Language-Action Models for OARbot Control

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1 LfD for Complex Tasks — VLA Control of OARbot

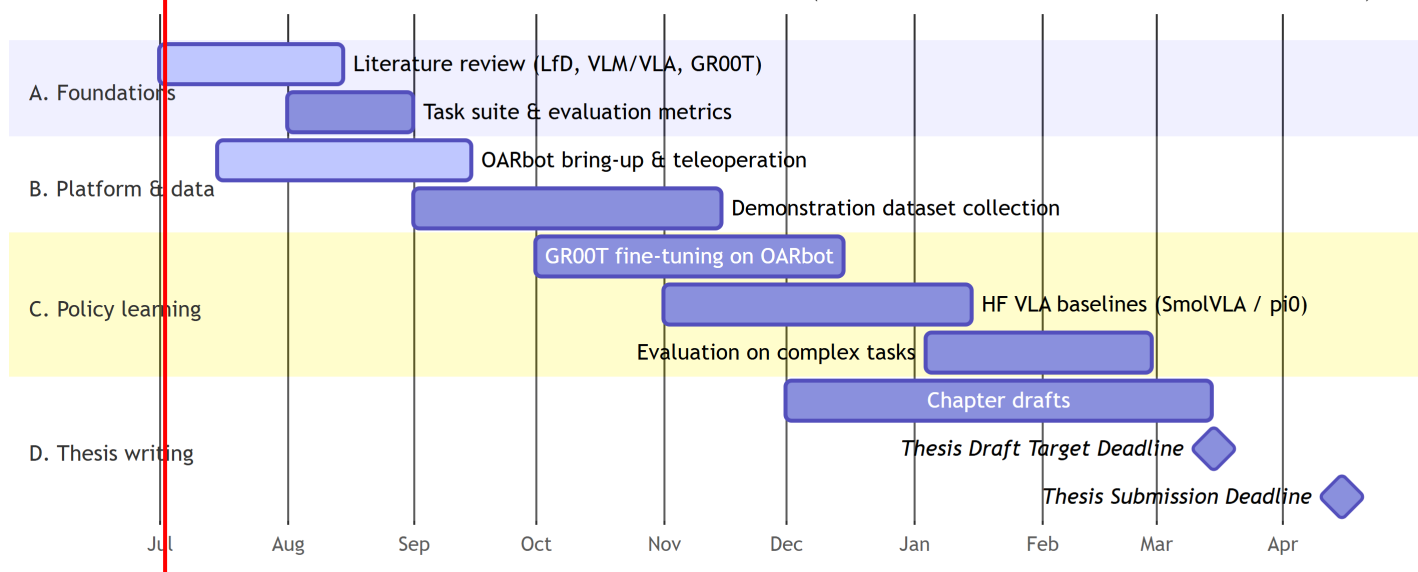
This site has two halves. The **thesis** is the prose in [Background](#), [Methods](#), [Operations](#), and [Future Work](#) — that’s what gets bound. The rest of the site is the **project-management surface** the thesis runs on: a top-level Gantt, the initiative board below, a weekly-update log, and an archive of source material.

TODO: replace this paragraph with the abstract
once the thesis is drafted.

1.1 Project schedule

Top-level Gantt covering the work categories the advisor tracks. The Mermaid block below is the single source of truth for the schedule. Edit this block when the schedule changes; the per-WU frozen PNGs (`updates/WU-*/media/gantt.png`) regenerate automatically from it via `scripts/render/build_gantt.py`. Add a bar here whenever you scaffold a new epic (`.\thesis.ps1 new epic ...`).

Thesis schedule — LfD for OARbot control (dates TODO: confirm with advisor)



1.2 Initiative board

The Gantt's bars are *epics*. Each epic has a README under [initiatives/](#) with a customer/need/benefit brief, related-epics graph, and an items listing. Everything below is generated from README front matter by `scripts/render/build_board.py` (a pre-render hook) — edit the front matter, never these tables. Schedule emphasis (crit bars) lives in the Mermaid Gantt above.

1.2.1 A — Foundations — LfD, VLM/VLA, and the task suite

Gantt	Epic	Status	Owner
A1	Literature review: LfD, VLM/VLA, GR00T	in-progress	Pol del Castillo
A2	Task suite & evaluation metrics	open	Pol del Castillo

1.2.2 B — Platform & data — OARbot bring-up and demonstrations

Gantt	Epic	Status	Owner
B1	OARbot bring-up & teleoperation	open	Pol del Castillo
B2	Demonstration dataset collection	open	Pol del Castillo

1.2.3 C — Policy learning — GR00T and HF VLA fine-tuning + evaluation

Gantt	Epic	Status	Owner
C1	GR00T fine-tuning on OARbot	open	Pol del Castillo
C2	HF VLA baselines (SmolVLA / pi0)	open	Pol del Castillo
C3	Evaluation on complex tasks	open	Pol del Castillo

1.3 Item board

Open 1

A1-EX-001

Which VLA backbones fit OARbot and the available compute?

A1 · Pol del Castillo

In progress 0

Blocked 0

Done 0

1.3.1 Stale items

in-progress items with no commit in 3+ weeks.

None — every in-progress item has recent activity.

1.4 Where to go from here

- **Reading the thesis** — start with [Background](#) and use the sidebar.
- **Reviewing weekly progress** — see the [weekly updates](#) appendix (newest first).
- **Browsing the project tree** — see the [initiatives](#) appendix; each initiative lists its epics, each epic lists its items + status log.
- **Looking up source material** — see the [archive](#) appendix, with an [external index](#) for files too large to carry in git.

Part I

Background

2 Background

TODO: Part intro — why this research area matters and what question the thesis answers.

3 Example section

TODO: First content section. Chapter subsection files follow the `CH-<N><x>-<slug>.qmd` naming convention and are registered in `_quarto.yml` under `book.chapters`. Cite sources with `@key` against `references.bib`.

Part II

Methods

4 Methods

TODO: Part intro — equipment, experimental setup, and how evidence flows from the lab into the **initiatives/** tree (every experiment in the PM layer that produces thesis-grade results gets written up here).

Part III

Operations

5 Operations

TODO: Part intro for operating procedures. SOPs live in `sops/` and render in three audience variants via Quarto profiles (`--profile sop-detailed | sop-checklist | sop-walkthrough`). Each SOP carries a formal revision block (rendered by the shortcode) plus an auto-generated git change history.

6 SOP-01: Example procedure

6.1 Purpose

TODO: One sentence on what this procedure accomplishes and when it applies. (This SOP exists so the template renders a working example of the revision block + profiles + change history — replace it with your first real procedure.)

6.2 Prerequisites

TODO

6.3 Verification

TODO

6.4 Change history

Owner: TODO Owner

Date	Author	Change
2026-06-12	AmmarBarbee17	Initial commit: Quarto thesis + project-management template

7 Future work

TODO: What the thesis opens up but does not close.
Future-shell epics in the **initiatives/** tree are good
source material here.

A Weekly updates

TODO: Weekly status slides shared with the advisor. Each entry is a standalone revealjs deck. Click any card to open the slides; reveal.js navigation (arrow keys, ESC for overview) applies.

B Initiatives

TODO: Initiatives are the top-level work categories that mirror the sections of the project Gantt. Each initiative groups its epics; each epic groups its experiments, issues, and features.

C Archive

Historical source-of-truth content imported from outside the repo (cloud drives, old proposals, vendor docs). **Large vendor binaries** and pre-hosted media already represented elsewhere are catalogued in [external-index.qmd](#) rather than copied.

C.1 Buckets

Buckets are filesystem dumps under `archive/` — there is no per-bucket rendered page. Use a local clone to browse contents; this table is the inventory. Files over ~25 MB get an external-index entry instead of a copy (git-lfs carries the rest — see `.gitattributes`).

Bucket	Description	Files
<i>(none yet)</i>	TODO: add a row per imported bucket	—

See [external-index.qmd](#) for files that stay at their original cloud paths.

D External archive index

Files referenced by the archive but **not copied into the repo** (too large for git-lfs to carry efficiently, or already represented elsewhere). They remain at their original cloud paths — record the absolute path so future-you can find them.

D.1 Always-external (vendor + heavy media)

What	Location
<i>(none yet)</i>	TODO: e.g. C:\Users\you\OneDrive – University\research\vendor- installers

D.2 Oversize files skipped from buckets (> 25 MB)

File	Size	Source
<i>(none yet)</i>	—	—